

RSS-based Localization of Multiple Unknown Transmitters through Particle Simulation

Philipp Schulz*, Norman Franchi*, and Gerhard Fettweis*

*Technische Universität Dresden, Dresden, Germany. Email: {philipp.schulz2, norman.franchi, fettweis}@tu-dresden.de

Abstract—The positions of wireless transmitters constitute valuable information for efficient dynamic radio resource planning, especially when resources are limited and communications is expected to be reliable and resilient. Moreover, in case there are unintended transmitters, i.e., jammers, the ability to localize them becomes even more important, in order to react and initiate appropriate countermeasures to finally recover operational mode. A joint design of transmitter sensing and radio resource utilization is of considerable benefit for dynamic network optimization especially in varying environments. In this paper, we propose a novel approach based on received signal strength (RSS) to iteratively localize one or multiple transmitters with a cooperative sensing network through particle simulation. By incorporating a transmit power estimation, the transmitters can be known or unknown. In particular, our RSS-based approach is widely independent from the specific wireless technology. Thereby, spatio-temporally correlated shadowing is considered. Our proposal is evaluated for accuracy and computation time considering a path loss model for an industrial scenario.

Index Terms—Localization, Cooperative Spectrum Sensing, Received Signal Strength, Resilience

I. INTRODUCTION

In the context of ultra-reliable low-latency communications (URLLC), reliability obviously is of key importance. However, for critical applications reliability alone may not be sufficient as they also require *resilience*. According to [1], resilience is "the ability to prepare and plan for, absorb, recover from, or more successfully adapt to actual or potential adverse events." Applied to URLLC applications, this means that systems not only need to be reliable against failures but also have to be able to recover from failures. In particular, these failures include external causes and threats, such as interference, which can be both accidental (e.g., coexisting technologies) or even intentional (i.e., jammers). To react on and mitigate jammers, they not only need to be detected but also localization is highly beneficial, as it is also for the regular transmitters (TXs) of the system, e.g., to monitor interference. Furthermore, a joint design of transmitter sensing and radio resource utilization provides invaluable input for dynamic network optimization especially in varying environments. In contrast to the regular TXs, jammers exhibit the disadvantage that they will not cooperate in any way to facilitate the localization.

Localization based on wireless communications technologies can be characterized into the following approaches: time-of-arrival (ToA) / time-difference-of-arrival (TDoA), angle-of-arrival (AoA), and received signal strength (RSS) [2]. ToA, TDoA, and AoA usually require known transmit signals as

prior knowledge. Furthermore, even for AoA based on RSS differences, it is not clear how AoA can be applied on multiple TX localization. Hence, an RSS approach is chosen here, which also offers low costs and easy implementation [3]. If TXs are known, an RSS approach is suitable for joint communications and localization. The RSS information is obtained from a network of distributed sensing units (SUs). For our approach it makes no difference, whether these SUs constitute a dedicated sensing network or whether elements of the actual network, e.g., access points (APs), serve as SUs. The provided information from the SUs has to be mapped to TX positions. This is commonly achieved by maximum likelihood estimation [3] or by solving the equations of an assumed path loss model in a least squares sense, e.g., in [4].

Existing approaches mostly refer to localization of *single* TXs with *known* transmit power and do not consider *correlated* shadowing. The work presented in [5] closes these gaps by proposing two algorithms for single TX localization (generalized least squares and maximum-likelihood-search) and two algorithms for multiple TX localization (radio environment map localization based on Kriging interpolation and compressed sensing (CS)) in scenarios with spatially correlated shadowing. This requires the knowledge of the stochastic parameters of the correlated shadowing. In [6], the same authors extended their CS approach to TX tracking with the help on an extended Kalman filter. However, for the tracking they do not consider that the shadowing may be correlated in time.

Our main contributions can be summarized as follows:

- 1) We provide a novel iterative approach for localization of one or multiple TXs based on particle simulation.
- 2) A transmit power estimation is incorporated to enable also the localization of unknown TXs.
- 3) Therein, we consider spatio-temporally correlated shadowing and potentially unknown transmit powers.
- 4) The proposed method is numerically evaluated with a path loss model from an industrial scenario with respect to accuracy and computation time.

II. SYSTEM MODEL

At first we describe the scenario as illustrated in Fig. 1. The considered area $\mathcal{R} \subset \mathbb{R}^2$ comprises a rectangular region with dimensions $L_w \times L_h$. In this area, M SUs are deployed. We assume that they are located at $M_x \times M_y$ equally spaced points $s_m \in \mathcal{R}$ on a rectangular grid, covering the entire region. It should be noted that any other shape of the region or any other deployment could be considered in our approach.

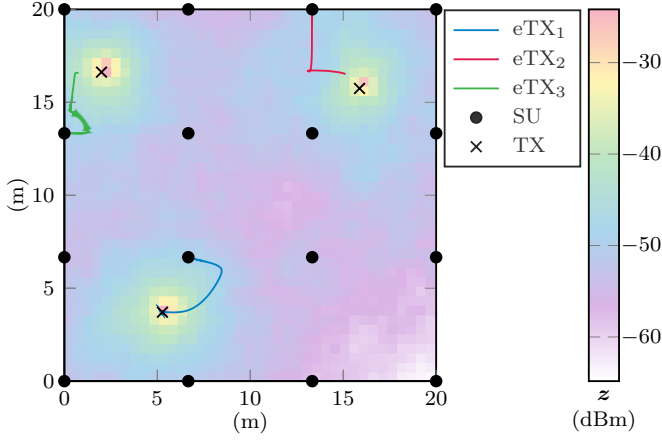


Fig. 1: Example Scenario. $M = 16$ SUs are deployed in a rectangular grid and $N = 3$ TXs are dropped randomly. In the background a simulated random realization of received powers with log-normal shadowing ($\sigma = 2$ dB) is indicated. Trajectories of the approach described in Sec. III are also included.

Within the region $\mathcal{R}$, we assume N TXs located at positions $\mathbf{u}_n \in \mathcal{R}$, which we aim to localize. The matrix $\mathbf{u} = [\mathbf{u}_1; \dots; \mathbf{u}_N]$ collects all TX coordinates. For this work, all TXs are assumed to be static. However, the approach is also expected to be suitable to track moving TXs. Each TX n transmits a signal with a transmit power $p_{\text{tx},n}$. In this work, isotropic antennas are assumed, such that the energy is distributed and received uniformly in each direction. The transmitted signals then experience the linear path loss $\chi(d)$ after traveling the distance d according to a two-slope model [7], i.e.,

$$\chi(d) = \begin{cases} \left(\frac{\lambda}{4\pi d}\right)^2, & d < d_0 \text{ (free space)} \\ \left(\frac{\lambda}{4\pi d_0}\right)^2 \left(\frac{d}{d_0}\right)^{-\gamma}, & d > d_0 \end{cases}. \quad (1)$$

Herein, λ refers to the wave length and γ denotes the path loss exponent. It should be noted that any other path loss model could be integrated into our approach to take different scenarios into account. With this and based on the TX positions $\mathbf{u}$, the SU m observes the following average receive power:

$$\bar{p}_{\text{rx},m}(\mathbf{u}) = \sum_{n=1}^N \chi(\|\mathbf{u}_n - \mathbf{s}_m\|_2) p_{\text{tx},n}. \quad (2)$$

In this paper, noise and fast fading effects are assumed to be averaged out and their impact left for future work. However, the slower fading induced by shadowing is taken into account. Here, correlated log-normal shadowing is considered, such that the experienced receive power in the log domain becomes:

$$\begin{aligned} 10 \log_{10}(p_{\text{rx},m}(\mathbf{u})) &= 10 \log_{10}(\bar{p}_{\text{rx},m}(\mathbf{u})) + w_m \\ z_m(\mathbf{u}) &= \bar{z}_m(\mathbf{u}) + w_m. \end{aligned} \quad (3)$$

Herein, the z variables refer to the received power in the log domain, i.e., in dB. There, the shadowing follows a multi-variate normal distribution, i.e., $\mathbf{w} = [w_1, \dots, w_M] \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with zero mean, $\boldsymbol{\mu} = \mathbf{0}$, and a distance-based correlation $\Sigma_{mn} = \sigma^2 \exp\left(-\frac{\|\mathbf{s}_m - \mathbf{s}_n\|_2}{d_{\text{corr}}}\right)$, as described in [8].

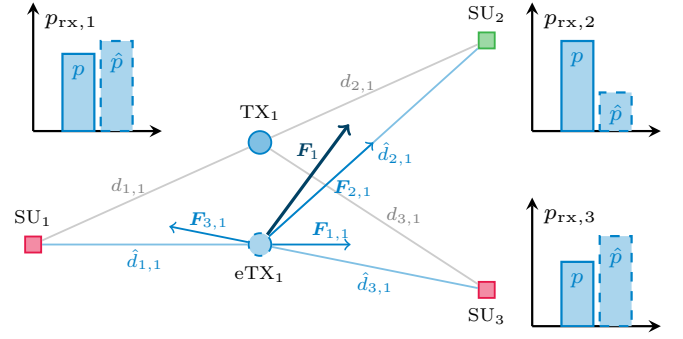


Fig. 2: Illustration of the proposed iterative localization approach.

The aforementioned equations can be extended for multiple time samples t_i , such that we have (3) at all time instances:

$$z_m^{(i)}(\mathbf{u}^{(i)}) = \bar{z}_m^{(i)}(\mathbf{u}^{(i)}) + w_m^{(i)} \quad (4)$$

To take into account that samples from same locations but different time instances do not observe independent shadowing, we introduce a correlation time t_{corr} , by extending $\mathbf{w}$ by components for each t_i . Accordingly, the entries in the covariance matrix $\boldsymbol{\Sigma}$ are extended by a time correlation factor:

$$\Sigma_{mn}^{(i,j)} = \sigma^2 \exp\left(-\frac{\|\mathbf{s}_m - \mathbf{s}_n\|_2}{d_{\text{corr}}}\right) \exp\left(-\frac{|t_i - t_j|}{t_{\text{corr}}}\right) \quad (5)$$

This approach includes the extreme cases of no time correlation ($t_{\text{corr}} = 0$) as well as full time correlation ($t_{\text{corr}} = \infty$). In general, the correlation time reflects a changing environment, e.g., moving obstacles, leading to shadowing effects. In static environments, the shadowing behavior would not change over time, leading to full time correlation.

III. LOCALIZATION APPROACH

First, only samples from one time instance are being considered. Multiple time samples will be handled in Sec. III-F.

A. Particle Simulation

The main idea comes from a physical interpretation. All SUs are considered as fixed particles. In addition, there are N estimated transmitters (eTXs) located at some positions $\hat{\mathbf{u}}_n \in \mathcal{R}$ which are considered as particles that are allowed to move during the iterations of a particle simulation. The initialization of the eTXs will be discussed in Sec. III-C. For now, it is also assumed that the transmit powers of the TXs are known. This assumption will be dropped in Sec. III-E.

The average receive power $\bar{p}_{\text{rx},m}(\hat{\mathbf{u}})$, which the SUs would receive from the TXs if they were at the eTXs locations, can be calculated according to (2). Here, the difference to the measured receive power (in log domain)

$$\Phi_m(\hat{\mathbf{u}}) = z_m(\mathbf{u}) - \bar{z}_m(\hat{\mathbf{u}}), \quad (6a)$$

$$= \bar{z}_m(\mathbf{u}) - \bar{z}_m(\hat{\mathbf{u}}) + w_m \quad (6b)$$

will be called the potential Φ_m at SU m . Instead of directly minimizing the localization error, which is not available, we try to minimize the total potential, i.e.,

$$\min_{\hat{\mathbf{u}} \in \mathcal{R}^N} \|\Phi(\hat{\mathbf{u}})\|_2. \quad (7)$$

A positive (negative) potential indicates that an SU measures a higher (lower) receive power than the one which would be expected from the eTXs. The vector Φ comprises all individual potentials. Based on this interpretation, we introduce a force $F_{n,m}$ from each SU m on each eTX n :

$$F_{n,m} = \Phi_m(\hat{\mathbf{u}}) \cdot \frac{\mathbf{s}_m - \hat{\mathbf{u}}_n}{\|\mathbf{s}_m - \hat{\mathbf{u}}_n\|_2^3}. \quad (8)$$

Therein, the second factor includes the normalized direction from the TX to the considered SU. The magnitude of this force is determined by the absolute value of the potential Φ_m and decays quadratically with the distance (like, e. g., gravitational force). Hence, the closer an SU to an eTX and/or the larger the potential is, the stronger is the induced force. Finally, the sign of the potential determines, whether an SU pulls eTXs towards itself or pushes them away, depending on whether they measure a too weak or a too strong receive power, respectively. Summing up all forces for one eTX n ,

$$\mathbf{F}_n = \sum_{m=1}^N \mathbf{F}_{n,m}, \quad (9)$$

the resultant force on each eTX can be calculated, as illustrated in Fig. 2. Let $F_{\max} = \max_n \{\|\mathbf{F}_n\|\}$ denote the strongest magnitude of these forces. It can be noted that without any random effects (like shadowing) the forces would be zero for $\hat{\mathbf{u}} = \mathbf{u}$, such that an equilibrium would be established. However, equilibria can also be established in points, where non-zero forces cancel each other out.

In contrast to real world particles, no mass or inertia is considered here to avoid overshooting. Instead, the movement in one time step is simply given by the force from (9):

$$\Delta_n^{[k]} = \begin{cases} \mathbf{F}_n^{[k]}, & F_{\max} \leq \Delta_{\max}, \\ \mathbf{F}_n^{[k]} \frac{\Delta_{\max}}{F_{\max}}, & F_{\max} > \Delta_{\max}. \end{cases} \quad (10)$$

However, the movement is limited to a maximum value $\Delta_{\max}$, since forces can become very large for eTXs being close to an SU. If any force exceeds this value, all forces are scaled down. In (10) an iteration index k was introduced, meaning that all required values (positions, forces) have to be recalculated for each iteration, in which the eTX positions are updated as:

$$\hat{\mathbf{u}}_n^{[k+1]} = \hat{\mathbf{u}}_n^{[k]} + h^{[k]} \Delta_n^{[k]}. \quad (11)$$

Herein, $h^{[k]}$ denotes the step size, which will be discussed in Sec. III-D. Even though (11) defines a particle simulation for several time steps, it should be noted that this time is purely virtual and does not coincide with the time in the real scenario. All iterations refer to a single time instance in the real world.

B. Exit Criteria

The iterations defined by (11) are subsequently carried out with the aim of minimizing the localization error. Accordingly, exit criteria need to be defined for the algorithm. A desired criterion would be a location error smaller than some tolerance. However, as this would require the true location, this criterion

would not be feasible. Also, no error estimator is available here, which could be used instead. Hence, the algorithm stops, if at least one of the following criteria is met:

$$k > k_{\text{break}}, \quad (12a)$$

$$\|\Phi\|_2 < \Phi_{\text{break}}, \text{ or} \quad (12b)$$

$$\sum_n \|\Delta_n\|_2 < \Delta_{\text{break}}. \quad (12c)$$

The algorithm stops if a maximum number k_{break} of iterations is exceeded (12a), the norm of the total potential Φ falls below a tolerance (12b), or if the total movement of all eTXs is lower than a predefined threshold (12c).

Inspecting (6b), it can be observed that for the desired solution $\hat{\mathbf{u}} = \mathbf{u}$, the remaining potential is given by the shadowing, i. e., $\Phi_0 = \mathbf{w}$. Hence, as the shadowing values are not known in general, a reasonable choice for Φ_{break} lies in the order of $\mathbb{E}[\|\mathbf{w}\|_2]$.

C. Initialization of the eTXs

The eTXs could be dropped just randomly into the scene to obtain initial values $\hat{\mathbf{u}}^{[0]}$ for the iteration in (11). However, an educated guess is preferred in order to finish the iterations faster. The eTXs are iteratively initialized at SU positions:

$$\hat{m}_1 := \operatorname{argmax}_m \Phi_m(\emptyset), \quad (13a)$$

$$\hat{m}_n := \operatorname{argmax}_m \Phi_m([\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_{n-1}]), \quad n > 1, \quad (13b)$$

$$\hat{\mathbf{u}}_n := \mathbf{s}_{\hat{m}_n}. \quad (13c)$$

The first eTX is put at the SU with the strongest experienced receive power (13a). As no eTX is considered in the potential $\Phi_m(\emptyset)$, this is equal to the measured received power at the SUs. For each following initialization of an eTX the potentials based on the already initialized eTXs are taken into account, and the index of the SU with the highest potential is chosen (13b). In other words, the SUs with the strongest remaining receive powers after subtracting expected powers from already placed eTXs are chosen. However, as receive powers cannot be negative, the notion is based on potentials instead.

D. Step Size Control

In numerics, the step size $h^{[k]}$ has a crucial role: small steps require more iterations, whereas large steps may overshoot the desired solution. Hence, an adaptive scheme is applied. Therefore, in each iteration, (11) is evaluated for the step sizes h , h_- , and h_+ , whereby the latter ones are defined by

$$h_+^{[k]} = \rho_+ \cdot h^{[k]}, \quad h_-^{[k]} = \rho_- \cdot h^{[k]}, \quad (14)$$

with $\rho_+ > 1$, $\rho_- < 1$ being parameters that specify the aggressiveness of the step size control. Afterwards, the step size that leads to the minimum total potential is chosen and will be the baseline $h^{[k+1]}$ for the next iteration.

E. Transmit Power Estimation

So far, a known transmit power was assumed for each TX to describe the approach. However, to enable jammer localization a transmit power estimation based on the current eTX positions and the path loss equation (2) is integrated into the approach.

1) *One Transmitter:* For a single TX only, the path loss (2) has a very simple representation in the log domain. With

$$L_m(\mathbf{u}_n) = 10 \log_{10} (\chi(\|\mathbf{u}_n - \mathbf{s}_m\|_2)), \quad (15a)$$

$$x_n = 10 \log_{10} (p_{\text{tx},n}), \quad (15b)$$

being the counterparts of the path loss and the transmit powers in the log domain, respectively, (2) can be written as

$$\bar{z}_m(\mathbf{u}_1) = x_1 + L_m(\hat{\mathbf{u}}_1). \quad (16)$$

Regarding all SUs and including the shadowing, the received powers in log domain become

$$\mathbf{z}(\mathbf{u}) = \mathbf{1}x_1 + \mathbf{L}(\hat{\mathbf{u}}_1) + \mathbf{w}. \quad (17)$$

As $\mathbf{w}$ is Gaussian, an optimal solution of this overdetermined system can be easily obtained by a least squares approach.

2) *Multiple Transmitters:* In the case of multiple eTXs, considering (2) in the log domain is not tractable anymore. Instead we reformulate (2) in matrix-vector notation by collecting all path loss coefficients in the path loss matrix $\mathbf{X}(\hat{\mathbf{u}})$

$$\bar{\mathbf{p}}_{\text{rx}}(\mathbf{u}) = \mathbf{X}(\hat{\mathbf{u}})\mathbf{p}_{\text{tx}}. \quad (18)$$

This overdetermined system can still be solved in a least squares sense. However, it should be noted that this approach is suboptimal, as shadowing is not Gaussian in the linear domain. Furthermore, the received powers differ over a range of orders of magnitude. Compared to an SU that experiences a strong received power, all SUs with weak received powers almost look the same, namely roughly zero. To take this into account for our least square estimation, we normalize the rows of (18) by defining $\mathbf{D} = \text{diag}(\bar{\mathbf{p}}_{\text{rx}}(\mathbf{u}))$:

$$\mathbf{1} = \mathbf{D}^{-1}\mathbf{X}(\hat{\mathbf{u}})\mathbf{p}_{\text{tx}}. \quad (19)$$

Solving (19) in a least square sense provides an estimate of $\mathbf{p}_{\text{tx}}$ in each iteration of the particle simulation.

F. Multiple Time Samples

This paper also aims to identify the potential of observing T time samples according to (4) in order to reduce the impact of shadowing. Here, we simply average the samples of receive powers at the SUs and are interested in the influence of time correlation. More sophisticated approaches, especially to enable tracking of moving TXs, are left for further studies.

IV. NUMERICAL EVALUATION AND VALIDATION

After introducing the scenario in Sec. IV-A, we will show a specific example based on a particular realization (Sec. IV-B). Afterwards, the performance will be evaluated statistically.

A. Scenario Parameters

For evaluation, we fix the area size to $L_w = L_h = L = 20$ m. As for the path loss, we consider a carrier frequency $f_c = 2.4$ GHz for which the path loss exponent $\gamma = 3.44$ was obtained in [7] with reference distance $d_0 = 15$ m. Each TX n has transmit power $p_{\text{tx},n} = 1$ mW. The correlation distance of the shadowing is set to $d_{\text{corr}} = 5$ m, which refers roughly to the maximum correlation distance obtained in [7]. Statistical metrics are always obtained from one hundred runs.

B. Numeric Example

In order to visualize the approach, an example run is provided. The results refer to the shadowing realization presented in Fig. 1, where also the eTX trajectories from the particle simulation are depicted. The eTXs are initialized at certain SUs (according to Sec. III-C) and move then to the desired TX locations. Interestingly, the initial direction is not necessarily towards the TX positions, but directions change due to the movements of the other eTXs and the updated potentials.

Several metrics for this particular realization are presented in Fig. 3. First, the development of the localization errors is shown in Fig. 3a. As there is an ambiguity for the mapping between eTXs and TXs, the mapping that minimizes the sum of all individual errors is chosen for the evaluation in each iteration. Hence, this mapping can also change during the simulation. However, for the localization the specific mapping is not of interest, as the eTXs are purely virtual. A decreasing trend of all localization errors can be observed.

For the break condition (cf. Sec. III-B) the algorithm needs to rely on other metrics, such as the (overall) eTX movement (Fig. 3b) or the total potential (Fig. 3c). It can be observed that the movement of eTX₃ is limited by $\Delta_{\text{max}} = 0.1$ m and after slightly more than 300 iterations it drops down. This happens at the same time, when the total potential Fig. 3c also drops down to the dashed line, which indicates the potential at the true positions due to the shadowing. Finally, Fig. 3d shows the development of the transmit power estimation. In the beginning, when all eTXs are close to the SUs, all powers are strongly under-estimated. However, for eTXs 1 and 2 the estimates increase fast until they reach approximately the true 1 mW. For eTX₃ this takes few more than 300 iterations, i. e., when the localization errors and other metrics also decline.

C. Localization Error and Transmit Power Estimation

The statistical results for single time samples¹ are depicted in Fig. 4. The plots in Fig. 4a and 4b show the mean individual localization errors for one and three TXs, respectively.

A higher number of TXs leads to higher individual localization errors, because TX localization is negatively affected by inaccurate locations of the other TXs. Furthermore, it can be observed that the localization error decreases for higher numbers of SUs in both cases. The standard deviation σ of the shadowing also has a strong impact on the localization error. Without any shadowing ($\sigma = 0$ dB), the approach is very accurate already for few SUs in the case of one TX, but also for three TXs, if there are 25 or more SUs. For higher values of σ , i. e., stronger shadowing effects, the accuracy decreases.

Interestingly, the transmit power estimation only has a small impact on the localization accuracy. In fact, the transmit power is well estimated for one TX (cf. Fig. 4c). In contrast, the transmit power estimation for three TXs experiences mean errors of 70 % for the strongest evaluated shadowing and 25 or more SUs due to the strong impact of localization errors

¹Single time samples still refer to an averaged received power over a short time to mitigate fast fading effects.

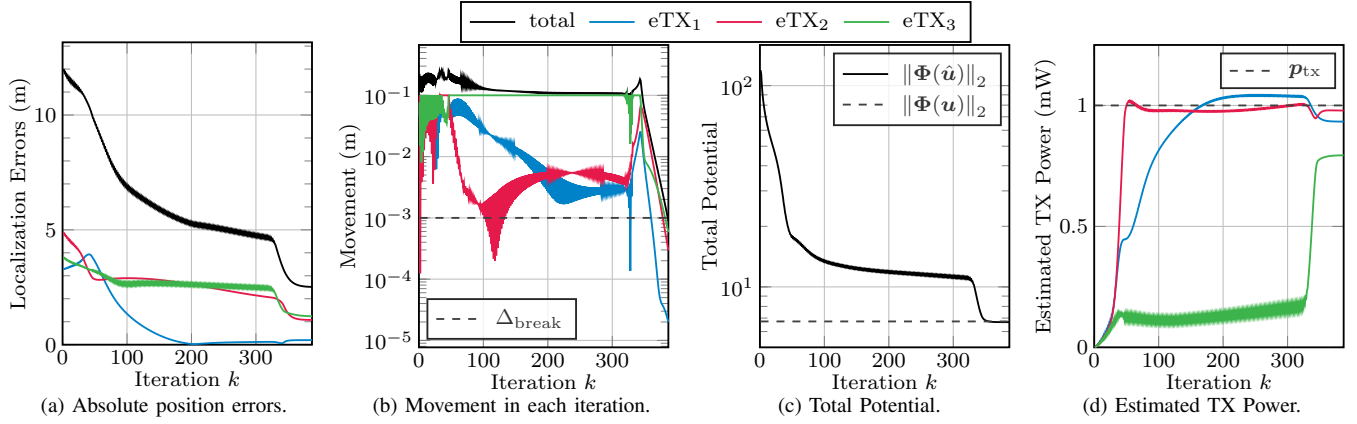


Fig. 3: Metrics over the iterations in a typical run for the example scenario. The legend on top applies to all subplots, i.e., black lines refer to all eTXs together, whereas colored lines belong to only one specific eTX.

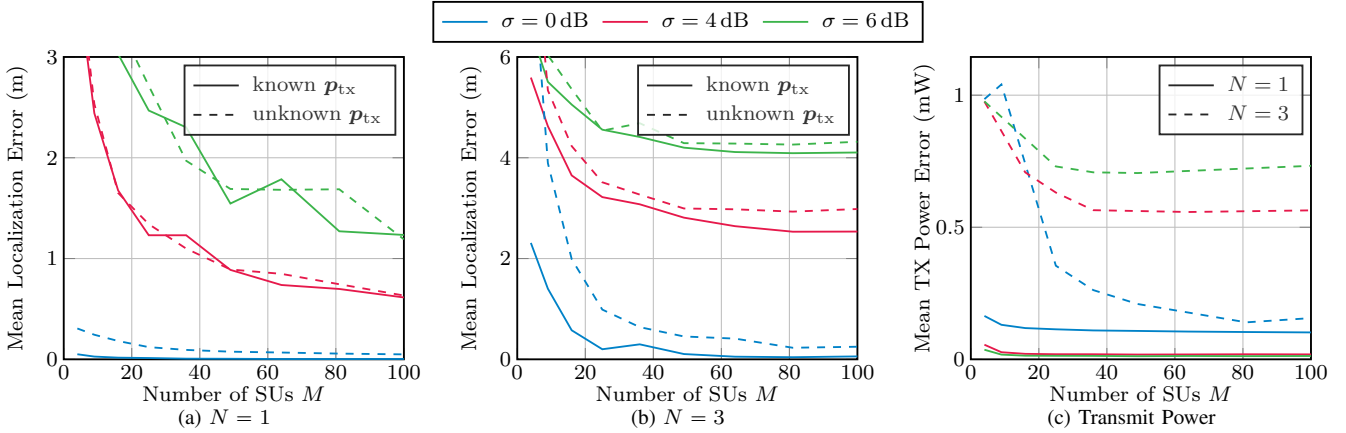


Fig. 4: Impact of transmit power estimation on localization performance and accuracy. The legend on top refers to all subplots.

on the path loss. The fact that the localization error is only slightly affected can be explained by looking at Fig. 2 again. If the transmit power is over-estimated (under-estimated), this affects all forces simultaneously, i.e., pushing forces become stronger (weaker) and pulling forces become weaker (stronger) compared to the forces that would result from the true value, leading to similar equilibrium points in between the SUs.

D. Multiple Time Samples and Time Correlation

The localization error can be significantly improved by observing more time samples to decrease the impact of the shadowing as shown in Fig. 5. As time correlation is assumed, results are obtained for different values of t_{corr} , which are normalized to the sample time Δt . It should be noted that the ratio $\frac{t_{corr}}{\Delta t} = 0$ refers to uncorrelated shadowing in time.

Figs. 5a and 5b show the results for one and three TXs, respectively. The same trends as for single time samples (cf. Fig. 4) can be observed. As expected, multiple time samples have no impact for the absence of shadowing ($\sigma = 0$). In contrast, the performance of single time samples (dotted lines) can be significantly improved through observing several samples. As anticipated, this gain is larger for less correlation.

In order to generate the plots in Fig. 5c/d, the number of SUs was fixed to $M = 81$ for one and three TXs, respectively.

This way, the impact of adding more time samples becomes evident. It can be observed that the localization error drops fast for the cases with no or weak correlation $\frac{t_{corr}}{\Delta t} \in \{0, 1\}$. For $\frac{t_{corr}}{\Delta t} = 10$ this process is approximately 10 times slower.

Regarding the performance assessment it should be noted that the TXs were dropped completely randomly, including cases where TXs are close to each other or close to the region border, making them harder to be localized. In contrast, e.g., [5] placed the TXs regularly spaced on a circle and did not consider time correlation between multiple time samples.

E. Computation Time

Lastly, the computational effort shall be studied. The run time mainly depends on the number of required iterations. Each iteration only involves comparably simple operations, i.e., calculating distances, path losses, forces and movements in the order of $\mathcal{O}(M \cdot N)$. Further analytical studies regarding the required number of iterations are left for future work.

Simulation results are depicted in Fig. 6 for one and three TXs, respectively. For $N = 1$ (Fig. 6a) a slightly increasing computation time can be observed for higher numbers of SUs, which can be explained due to more complex evaluations of each iteration. In contrast, a high peak can be observed for $N = 3$ (Fig. 6b) at low numbers of SUs, when transmit powers

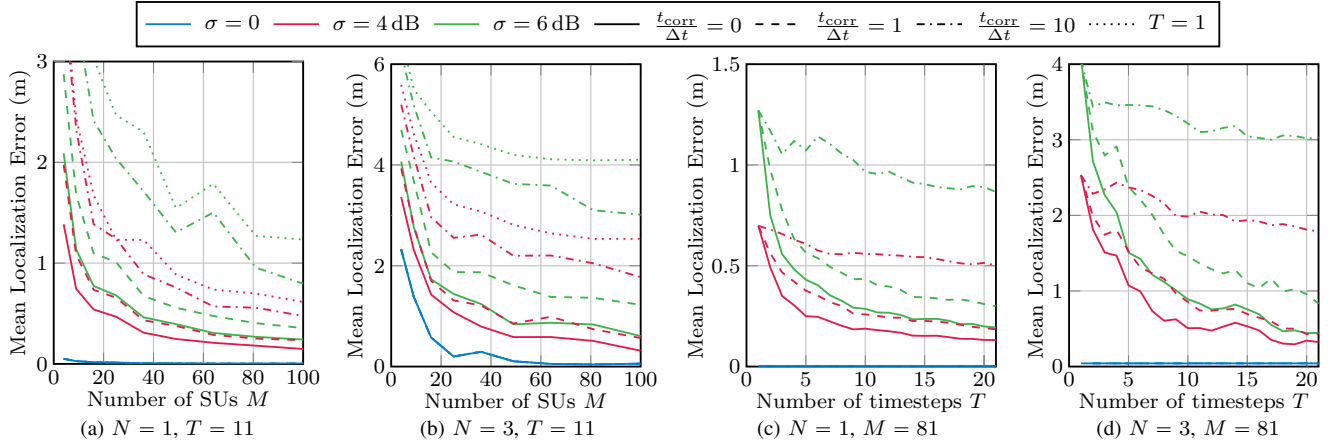


Fig. 5: Performance for multiple time samples T and varying time correlation. The legend on top applies to all plots.

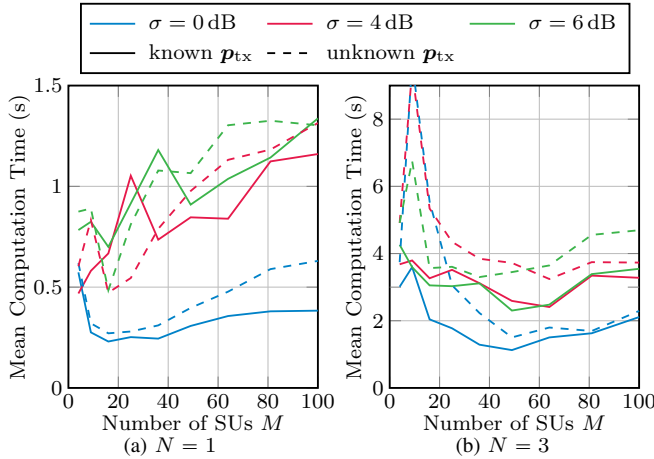


Fig. 6: Computation time with and without transmit power estimation for different values of the shadowing standard deviation σ .

have to be estimated. Then, potentially wrong estimates in the beginning seem to slow down the algorithm. Apart from that, the computation time is almost constant. This can be explained by the fact that though iterations are slightly more complex for more SUs, less are required for a higher number of SUs in this case. As expected, the transmit power estimation increases the required computation time in most cases.

Computation times in the order of a second (around three to four seconds for three TXs) are not yet suitable for real-time monitoring. However, this evaluation was not optimized for computation time yet. Once initialized, only few more iterations are expected for tracking tasks.

V. CONCLUSION AND OUTLOOK

In this paper, we proposed an iterative localization approach, solely based on received signal strength (RSS) measurements and evaluated it based on simulations. Here, spatio-temporally correlated fading was considered as a key error source. The approach is suitable for tracking tasks by design, as the algorithm can be initialized with the last known position and only few more iterations are expected for new positions. Our method provides a valuable tool towards joint design of

transmitter sensing and radio resource utilization and, thus, also towards network optimization in varying environments.

This paper only introduces a new localization approach offering great potential for future work. Possible topics include: theoretical analysis including convergence behavior, tracking, estimating the number of transmitters (TXs), directional antennas, real-world testing, combination with machine learning, etc. Furthermore, accuracy improvements are expected by introducing means for shadowing estimation. Finally, step-size control can be optimized and particle simulation methods from physics may be applied.

ACKNOWLEDGMENT

This work was funded and supported by the project “Industrial Radio Lab Germany” under contract 16KIS1010K, funded by the Federal Ministry of Education and Research, Germany. We also thank the anonymous reviewers for their valuable comments.

REFERENCES

- [1] *Disaster Resilience: A National Imperative*. Washington, DC: The National Academies Press, 2012. doi: 10.17226/13457
- [2] N. Patwari, J. N. Ash, S. Kyperountas, A. O. Hero, R. L. Moses, and N. S. Correal, “Locating the nodes: cooperative localization in wireless sensor networks,” *IEEE Signal Process. Mag.*, vol. 22, no. 4, pp. 54–69, 2005. doi: 10.1109/MSP.2005.1458287
- [3] R. W. Ouyang, A. K. Wong, and C. Lea, “Received signal strength-based wireless localization via semidefinite programming: Noncooperative and cooperative schemes,” *IEEE Trans. Veh. Technol.*, vol. 59, no. 3, pp. 1307–1318, 2010. doi: 10.1109/TVT.2010.2040096
- [4] K. W. Cheung, H. C. So, W.-K. Ma, and Y. T. Chan, “A constrained least squares approach to mobile positioning: Algorithms and optimality,” *EURASIP J. Adv. Signal Process.*, vol. 2006, no. 1, Dec 2006. doi: 10.1155/ASP/2006/20858
- [5] K. Kulkarni, N. Franchi, and G. Fettweis, “Reliable real-time localization and tracking of interferers using cooperative spectrum sensing,” in *IEEE Wireless Communications and Networking Conference (WCNC)*, Marrakech, Morocco, Apr 2019. doi: 10.1109/WCNC.2019.8885898
- [6] K. Kulkarni, N. Franchi, and G. Fettweis, “Real-time cooperative spectrum sensing: Reliable localization of unknown interferers in shared spectrum,” in *International ITG Conference on Systems, Communications and Coding (SCC)*, Rostock, Germany, Feb 2019. doi: 10.30420/454862016
- [7] E. Tanghe *et al.*, “The industrial indoor channel: large-scale and temporal fading at 900, 2400, and 5200 MHz,” *IEEE Trans. Wireless Commun.*, vol. 7, no. 7, pp. 2740–2751, 2008. doi: 10.1109/TWC.2008.070143
- [8] S. S. Szyszkowicz, H. Yanikomeroglu, and J. S. Thompson, “On the feasibility of wireless shadowing correlation models,” *IEEE Trans. Veh. Technol.*, vol. 59, no. 9, pp. 4222–4236, 2010. doi: 10.1109/TVT.2010.2082006